



SEMESTER END EXAMINATIONS – FEBRUARY 2024

Program : **B.E :- Information Science and Engineering**
Course Name : **Artificial Intelligence**
Course Code : **IS52**

Semester : **V**
Max. Marks : **100**
Duration : **3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

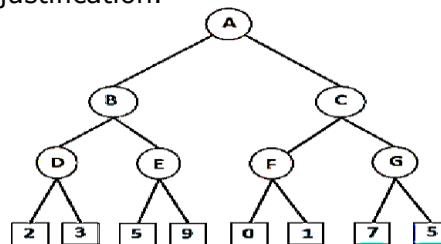
- Explain the following two categories of Artificial Intelligence definition with respect to thought processes and reasoning:
i) Thinking Rationally ii) Thinking Humanly. CO1 (06)
 - With a suitable diagram and example, describe the agents interaction with environment. CO1 (09)
 - Mention the PEAS description for an Automated taxi. CO1 (05)
- Explain the following properties of task environment:
i) Deterministic vs Stochastic ii) Single agent vs Multiagent. CO1 (08)
 - Explain how A* Search algorithm minimizes the total estimated solution cost. CO1 (08)
 - Solve the following 8 puzzle problem also write solution space tree. CO1 (04)

1	2	3
	4	6
7	5	8

1	2	3
4	5	6
7	8	

UNIT - II

- For the given figure, apply alpha-beta pruning. Clearly mention the pruned nodes with justification. CO2 (06)



- Explain entailment with an example for the WUMPUS world. CO2 (07)
 - Explain the syntax and semantics using prepositional logic for the Wumpus world. CO2 (07)
- Justify how alpha-beta pruning is more efficient than minimax strategy in reaching the goal state. Explain with an example. CO2 (10)
 - Explain the general process of knowledge base construction. CO2 (10)

UNIT - III

- Are the following two sentences equivalent? Justify our answer. CO3 (06)
 $\forall x \neg Likes(x, Parsnips)$
 $\neg \exists x Likes(x, Parsnips)$
 - Explain with examples the purpose of unification and lifting operations in first-order logic. CO3 (07)

- c) Generate the planning graph up to level S_2 for the following "have cake and eat cake too" problem: CO3 (07)

$Init(Have(Cake))$
 $Goal(Have(Cake) \wedge Eaten(Cake))$
 $Action(Eat(Cake))$
 PRECOND: $Have(Cake)$
 EFFECT: $\neg Have(Cake) \wedge Eaten(Cake)$
 $Action(Bake(Cake))$
 PRECOND: $\neg Have(Cake)$
 EFFECT: $Have(Cake)$

6. a) Explain simple backward chaining algorithm with an example. CO3 (08)
 b) Convert the following first order logic to conjunctive normal form: CO3 (07)
 $\forall x [\forall y Animal(y) \Rightarrow Loves(x, y)] \Rightarrow [\exists y Loves(y, x)]$.
 c) Describe the properties of planning graphs which are monotonically increasing or decreasing. CO3 (05)

UNIT- IV

7. a) Given the full joint distribution as shown in fig below, calculate the following. CO4 (08)
 i) $P(toothache)$
 ii) $P(Cavity)$
 iii) $P(Toothache|cavity)$
 iv) $P(Cavity|toothache \vee catch)$

	Toothache		$\neg$ Toothache	
	catch	$\neg$ catch	catch	$\neg$ catch
Cavity	0.108	0.012	0.072	0.008
$\neg$ Cavity	0.016	0.064	0.144	0.576

- b) Explain how objects are organized into categories, being the vital part of Knowledge representation. CO4 (06)
 c) Justify how an agent forms rational decisions in an uncertain environment. CO4 (06)
8. a) Consider two medical tests, A and B, for a virus. Test A is 95% effective at recognizing the virus when it is present but has a 10% false positive rate. Test B is 90% effective at recognizing the virus but has a 5% false positive rate. The two tests use independent methods of identifying the virus. The virus is carried by 1% of all people. Say that the person is tested for the virus using only one of the tests, and the tests come back positive for carrying the virus. Which test returning positive is more indicative of someone really carrying the virus? Justify the answer mathematically. CO4 (10)
 b) Explain Event Calculus and the need for Event Calculus in Ontological Engineering. CO4 (10)

UNIT - V

9. a) Illustrate how the squares *known*, *frontier* and *other* influences on handling uncertainty in wumpus world problem. CO5 (06)
 b) Describe three layer architecture of robot. CO5 (06)
 c) Discuss any four application domains of robotic technology. CO5 (08)
10. a) Describe different kinds of sensors used in robots. CO5 (08)
 b) Explain how robot handles uncertain movement using fine-motion planning. CO5 (07)
 c) Illustrate the situation when the reactive control is required for the robot. CO5 (05)
